Industrial Internship Report on

Multiclient marketplace platform

Prepared by

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Executive Summary
This report presents the work completed during my industrial internship, where I developed a Multi-Client Service Marketplace Platform. The system enables users to browse services, register, login, and book services, while service providers can manage their offerings. The project was developed using Java (Servlets & JSP), JDBC, MySQL, and MVC architecture. It focuses on building a scalable backend system with proper authentication, session management, and database integration. This internship provided hands-on experience in real-world application development, backend design, and problem-solving in a structured environment

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1 Preface

This internship was a valuable opportunity to gain practical exposure to real-world software development. Over a span of 6 weeks, I worked on designing and developing a service marketplace platform. Throughout this process, I gained a deeper understanding of backend development, including concepts such as Java Servlets, JSP, database connectivity using JDBC, and MVC architecture. I also learned how to manage user sessions, handle errors, and build a scalable and maintainable system.

The program helped me understand backend development, database design, and system architecture. I would like to thank my mentors and peers for their guidance and support throughout this journey.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

UniConverge Technologies Pvt Ltd (UCT) is a technology-driven organization established in 2013, focusing on digital transformation solutions for industries. The company specializes in delivering scalable and efficient solutions using modern technologies such as Internet of Things (IoT), cloud computing, machine learning, and full-stack development.

UCT provides innovative platforms that help industries improve operational efficiency, monitor performance, and make data-driven decisions. Their solutions are widely used in domains such as smart manufacturing, predictive maintenance, smart cities, and industrial automation. By leveraging technologies like Java, ReactJS, and cloud platforms such as AWS and Azure, UCT builds reliable and scalable systems for enterprise applications.

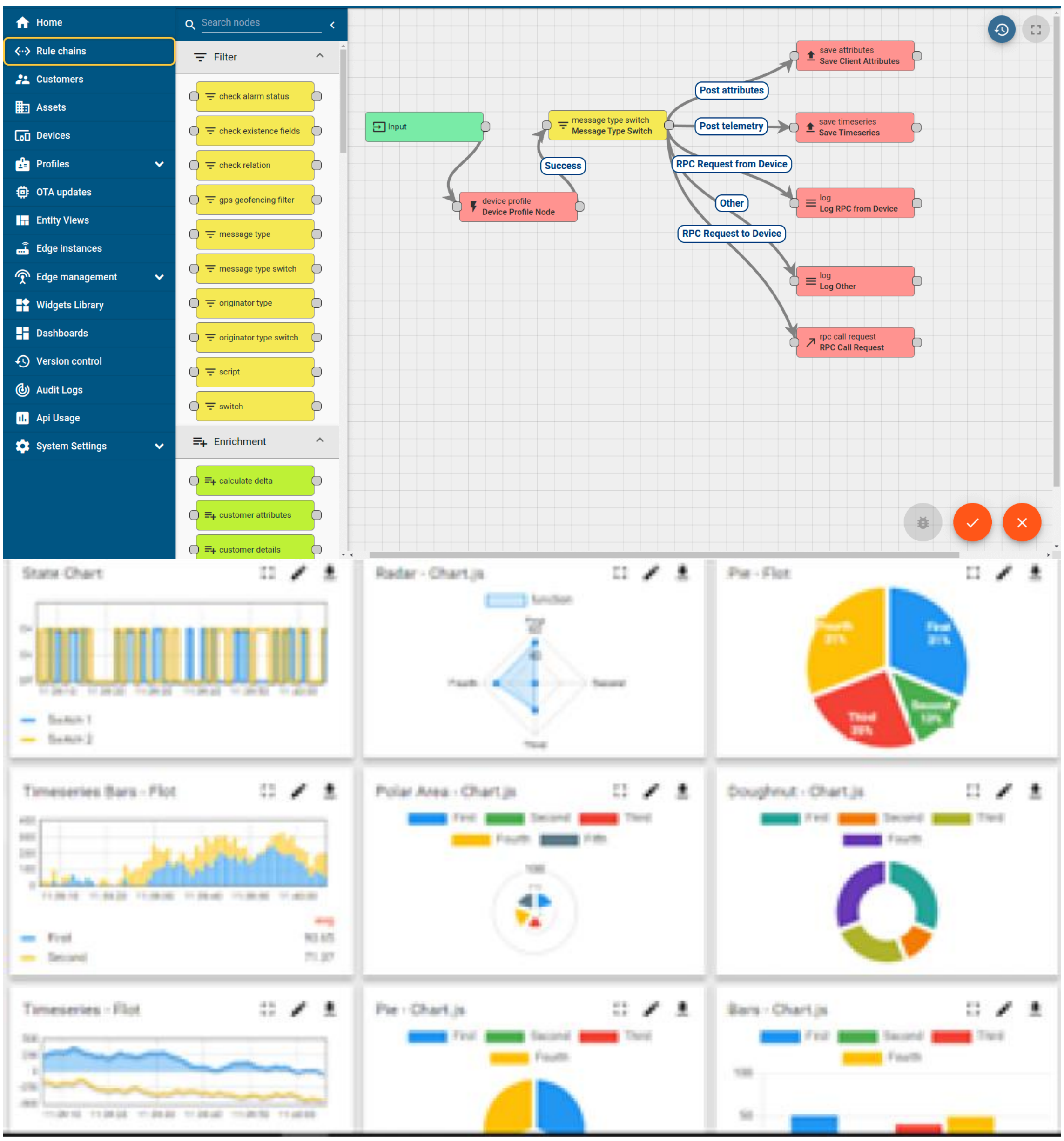
i. UCT IoT Platform (uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA. It supports both cloud and on-premises deployments.

It has features to

• Build Your own dashboard
• Analytics and Reporting
• Alert and Notification
• Integration with third party application(Power BI, SAP, ERP)
• Rule Engine



ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

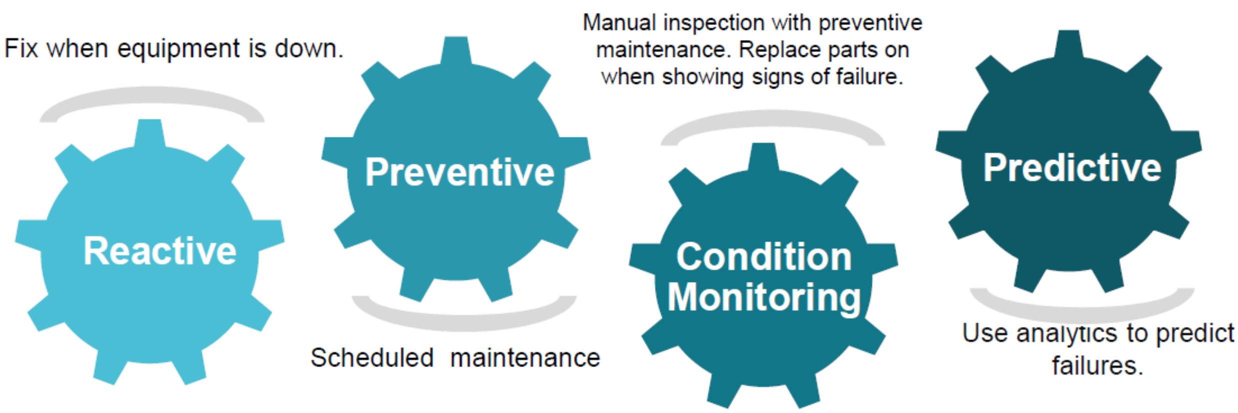
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

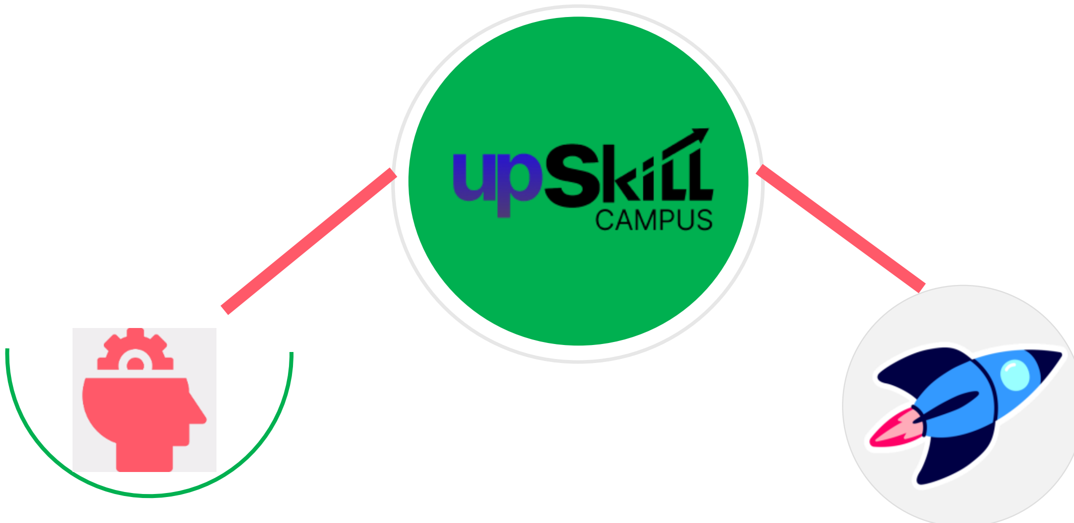
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

Upskill Campus (USC) is a career development platform that aims to bridge the gap between academic learning and industry requirements. It provides structured internship programs, live projects, mentorship, and career guidance to help students gain practical exposure.

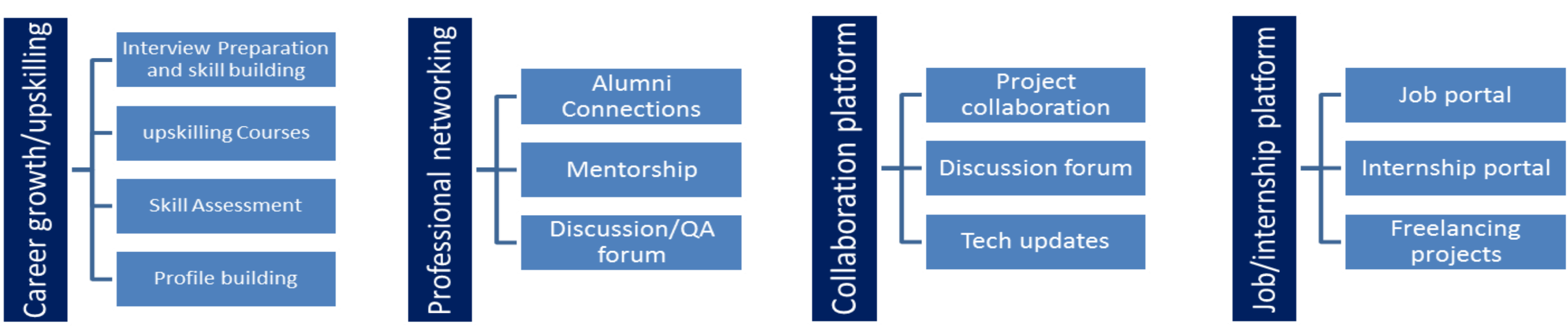
Through collaborations with industry partners like UCT, Upskill Campus enables students to work on real-world problem statements. The platform focuses on skill enhancement in areas such as software development, data science, and emerging technologies, preparing students for better career opportunities.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ To gain practical experience in software development

☛ To understand real-world problem-solving approaches

☛ To develop a complete end-to-end project

☛ To improve technical skills in backend development

☛ To enhance logical thinking and debugging skills

☛ To understand system design and architecture

2.5 Reference

[1] Java Documentation (Oracle)

[2] MySQL Documentation

[3] Apache Tomcat Documentation

2.6 Glossary

Terms	Acronym
MVC	Model View Controller architecture
JDBC	Java Database Connectivity
Servlet	Java program handling HTTP requests
JSP	Java Server Pages for UI
CRUD	Create, Read, Update, Delete operations

3 Problem Statement

The objective of this project is to develop a **Multi-Client Service Marketplace Platform** that connects service providers with customers. The system should allow users to browse available services, register/login securely, and book services efficiently.

The platform must handle multiple users and service providers simultaneously while ensuring proper data management and system scalability.

Key challenges include:

- Designing a scalable database structure
- Implementing secure authentication and session handling
- Managing multiple service providers and users
- Ensuring smooth booking functionality
- Maintaining data consistency and integrity

4 Existing and Proposed solution

Existing platforms such as Urban Company provide similar services where users can book professionals for various tasks. However, these platforms are large-scale systems with complex architectures.

- **Limitations**
- High complexity
- Not suitable for learning core concepts
- Limited customization
- Requires advanced infrastructure
- **Proposed Solution**

The proposed system is a simplified service marketplace platform designed to demonstrate core backend development concepts. It focuses on:

- User authentication system
- Service listing and management
- Booking functionality
- Database integration
- MVC-based architecture

This solution provides a clear understanding of how real-world applications are built.

4.1 Code submission(Github link)

https://github.com/ambikaaswal/upskillcampus

NOTE: The database password is set to "". Please update it according to your local MySQL configuration before running the project.

4.2 Report submission (Github link) :

https://github.com/ambikaaswal/upskillcampus/blob/main/MulticlientServiceMarketplacePlatform_AmbikaAswal_USC_UCT.pdf

5 Proposed Design/ Model

The system is designed using the **MVC architecture**, ensuring separation of concerns.

Working Flow

1. User registers or logs in
2. User views available services
3. User selects and books a service
4. Booking details are stored in the database
5. Service providers can manage services

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5.1 High Level Diagram (if applicable)

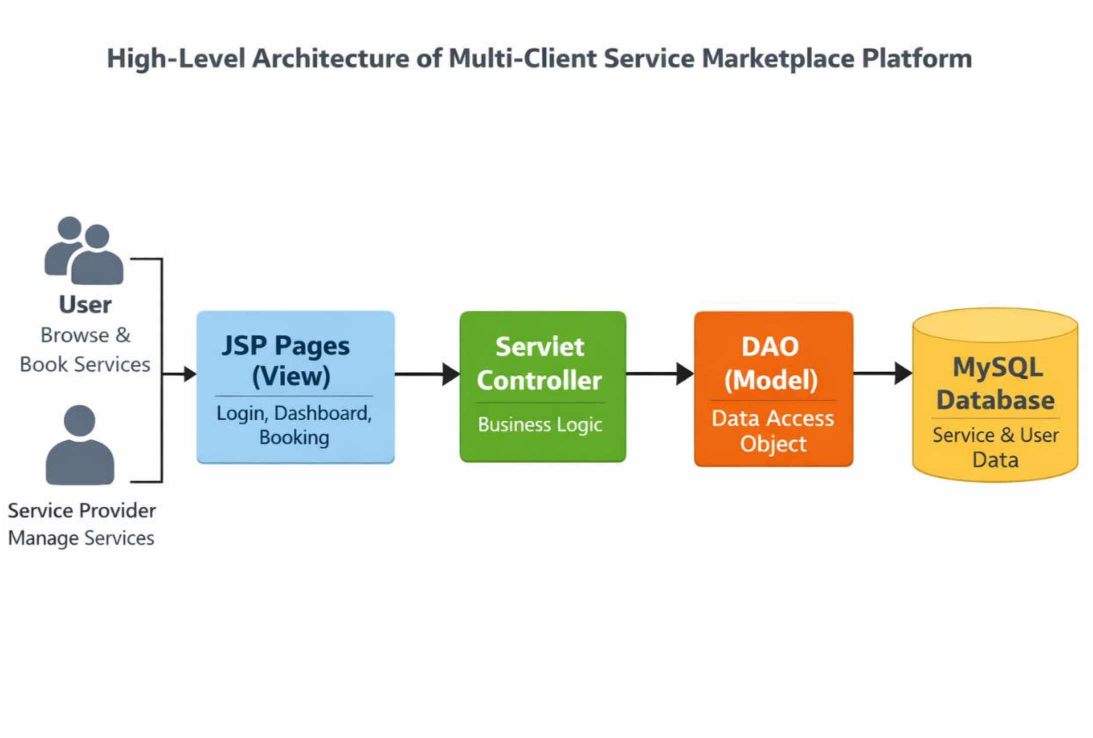
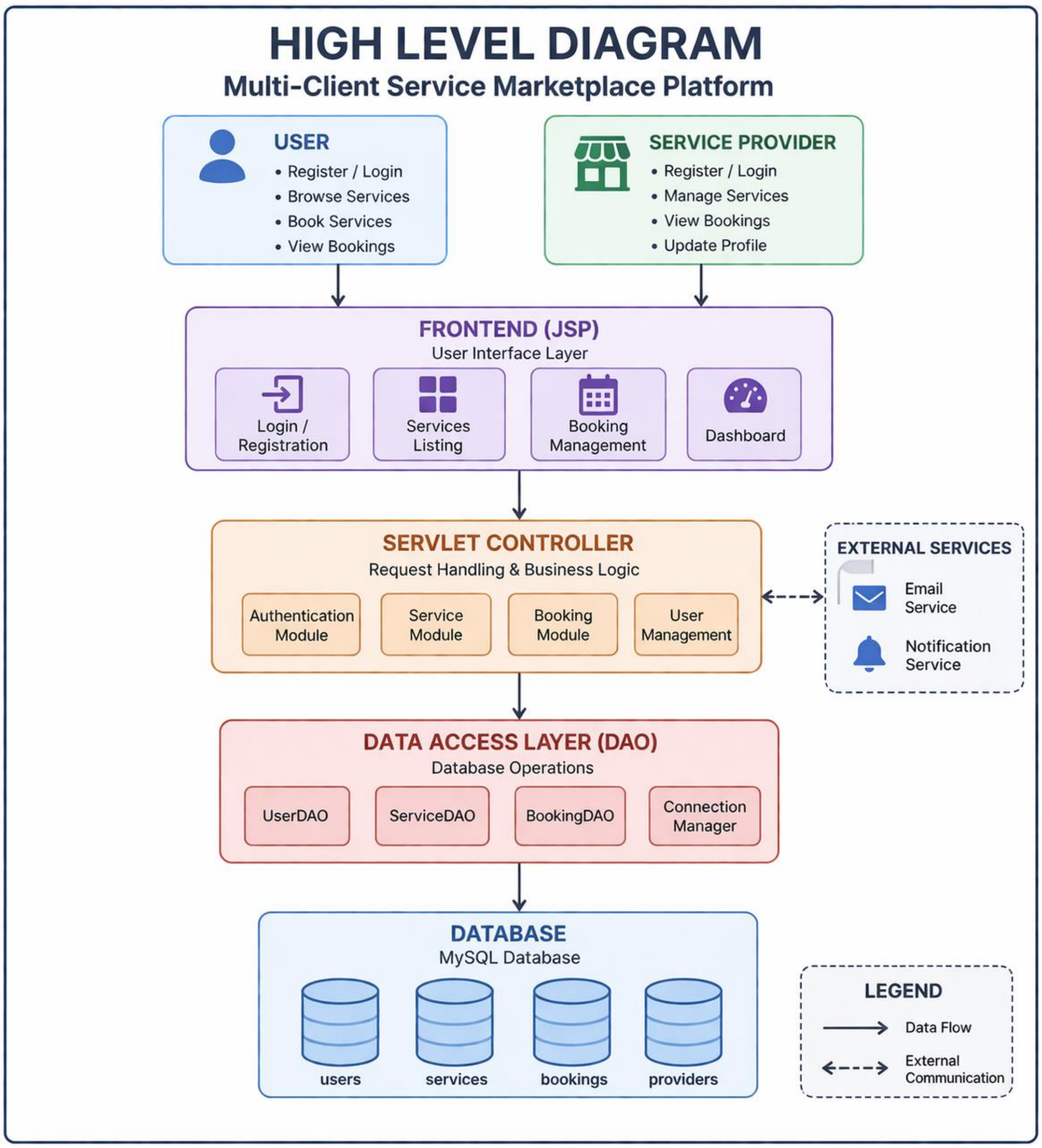
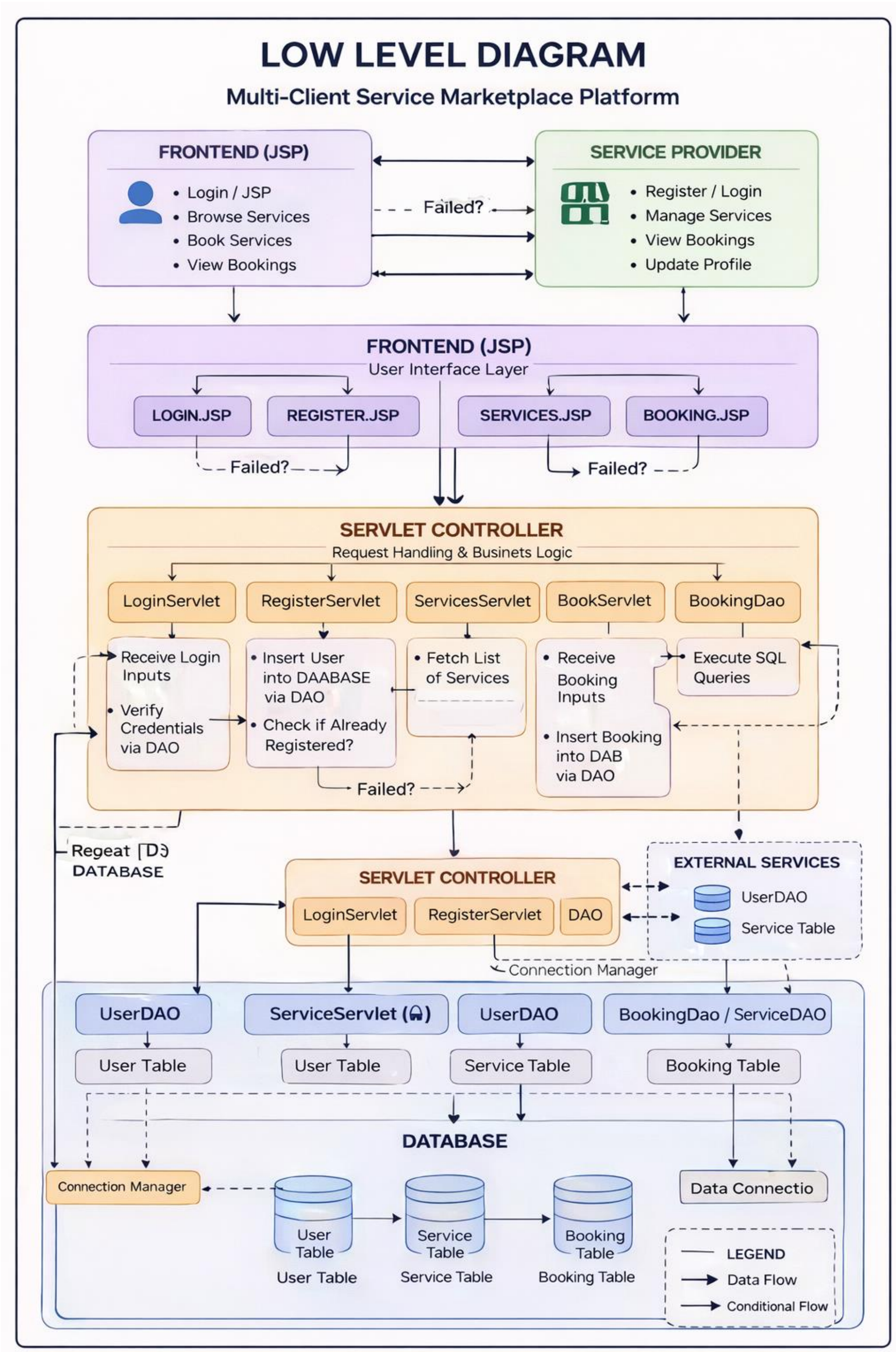


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM


HIGH LEVEL DIAGRAM
Multi-Client Service Marketplace Platform
USER
• Register / Login
• Browse Services
• Book Services
• View Bookings
SERVICE PROVIDER
• Register / Login
• Manage Services
• View Bookings
• Update Profile
FRONTEND (JSP)
User Interface Layer
Login / Registration
Services Listing
Booking Management
Dashboard
SERVLET CONTROLLER
Request Handling & Business Logic
Authentication Module
Service Module
Booking Module
User Management
EXTERNAL SERVICES
Email Service
Notification Service
DATA ACCESS LAYER (DAO)
Database Operations
UserDAO
ServiceDAO
BookingDAO
Connection Manager
DATABASE
MySQL Database
users
services
bookings
providers
LEGEND
Data Flow
External Communication

5.2 Low Level Diagram



5.3 Interfaces

In the Multi-Client Service Marketplace Platform, interfaces are designed using a layered architecture consisting of JSP (View), Servlets (Controller), and DAO (Model).

These interfaces ensure smooth communication between frontend, backend, and database layers.

GUI Interfaces of the project:

1. Login Page
2. Registration Page
3. Dashboard Page
4. Services Page
5. Booking Page

5.3.i Block Diagram Explanation

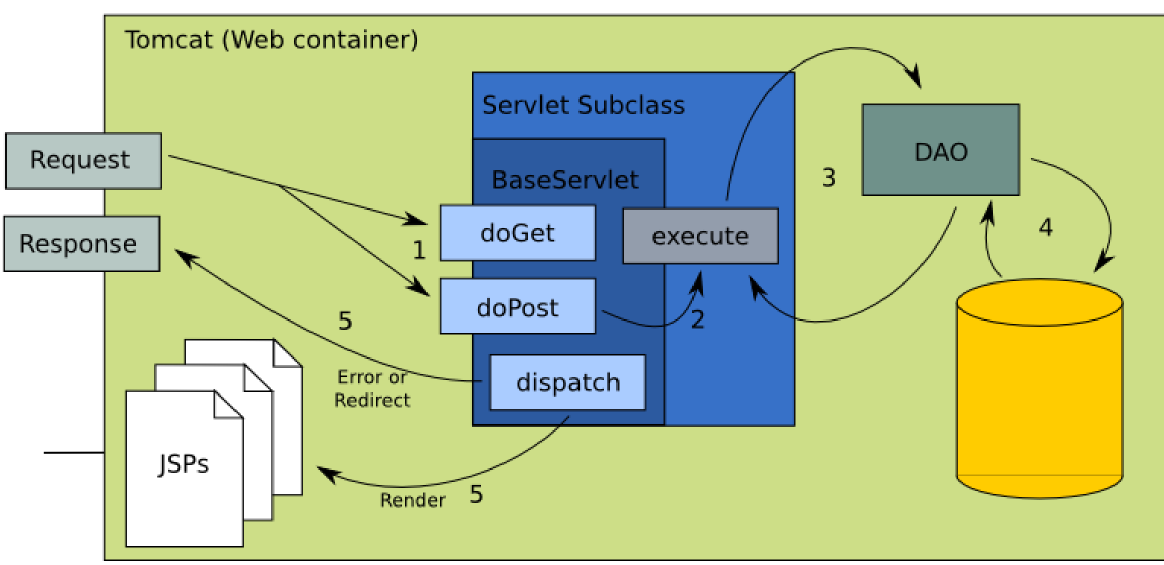
The system follows a three-tier architecture:

1. Presentation Layer (JSP): Handles user interface (login, dashboard, booking pages)

2. Business Logic Layer (Servlets): Processes requests and applies logic

3. Data Layer (DAO + MySQL): Handles database operations

Data flows from User → JSP → Servlet → DAO → Database



5.3.ii Data Flow Description

Booking Flow

1. User selects a service on services.jsp
2. Request is sent to BookServlet
3. Servlet processes booking logic
4. DAO stores booking in database
5. Response is sent back to JSP
6. User sees confirmation

This ensures structured request-response handling

5.3.iii Protocols Used

The system uses standard web communication protocols:

1. HTTP (HyperText Transfer Protocol)
   Used for communication between client (browser) and server
2. JDBC (Java Database Connectivity)
   Used for communication between Java application and MySQL database

6 Performance Test

6.1 Test Plan/ Test Cases

Test Case	Description
Login Test	Verify user login functionality
Registration Test	Verify new user creation
Booking Test	Check service booking
Database Test	Verify data storage

6.2 Test Procedure

1. Run application on Apache Tomcat
2. Perform user actions (login, booking, etc.)
3. Verify database entries
4. Check for errors

6.3 Performance Outcome

The system performed efficiently under normal usage conditions. All functionalities such as login, service display, and booking worked correctly. Data was stored and retrieved successfully from the database.

7 My learnings

During this internship, I gained significant knowledge and practical experience in software development. I learned how to build a complete backend system using Java technologies and understood the importance of structured development.

Key learnings include:

1. Understanding MVC architecture
2. Working with Servlets and JSP
3. Database integration using JDBC
4. Handling user sessions and authentication
5. Debugging errors and improving code quality
6. Designing scalable applications

This experience has strengthened my foundation in backend development and prepared me for real-world projects.

8 Future work scope

The project can be further enhanced with the following features:

1. Payment gateway integration
2. Real-time notifications and alerts
3. Improved UI using modern frameworks like React
4. Mobile application development
5. AI-based service recommendation system
6. Enhanced security features